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Assignment Number:14.3(Present assignment number)/**24**(Total number of assignments)

Lab 14: Web Design Applications Using AI Assistance

HTML, CSS, and JavaScript Generation with GitHub Copilot

Lab Objectives

The objectives of this laboratory exercise are to:

- Design functional and visually appealing web applications using HTML, CSS, and JavaScript.
- Use AI-assisted tools to accelerate front-end development.
- Apply responsive, accessible, and interactive web design principles.
- Customize and validate AI-generated UI code for real-world applications.

Learning Outcomes

After completing this lab, students will be able to:

- Generate structured HTML layouts using AI assistance.
- Apply responsive CSS techniques such as Flexbox and Grid.
- Implement interactive client-side features using JavaScript.
- Critically review and improve AI-generated front-end code.

Tools Required

- Visual Studio Code (VS Code)
- GitHub Copilot Extension
- Web Browser (Chrome / Firefox)

HTML5, CSS3, JavaScript (ES6

Task 1: AI-Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections:
    - Home
    - About
    - Projects
    - Contact
2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.
3. Customize AI-generated CSS to add:
    - Hover effects on project cards
    - Smooth scrolling navigation

Expected Outcome

- Clean and well-structured portfolio website
- Responsive layout for desktop and mobile screens
- Interactive project cards

Prompt: Generate a clean and responsive personal portfolio website using HTML and CSS.

Requirements:

- Use semantic HTML with sections: Home, About, Projects, Contact, and Footer

- Add a navigation bar with smooth scrolling links

- Use Flexbox for navbar and layout, and CSS Grid for project cards

- Create 3 project cards with title and description

- Add hover effects on project cards (lift + shadow + smooth transition)

- Make the navbar sticky

- Ensure full responsiveness for mobile (max-width: 768px)

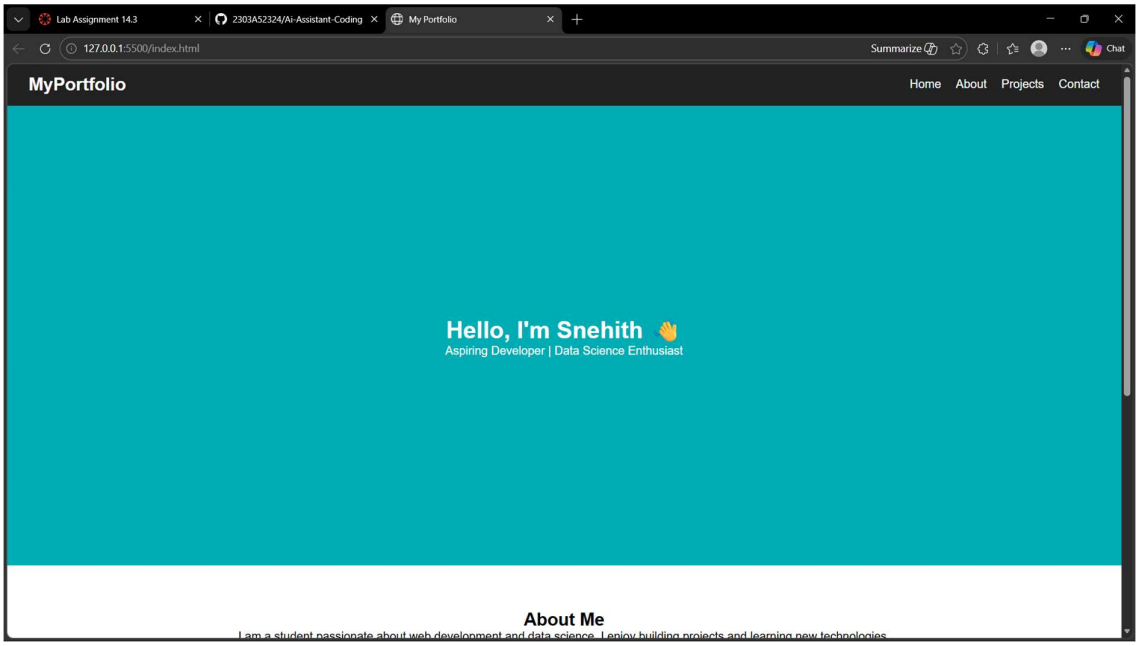
- Use a modern, minimal design with good spacing and typography

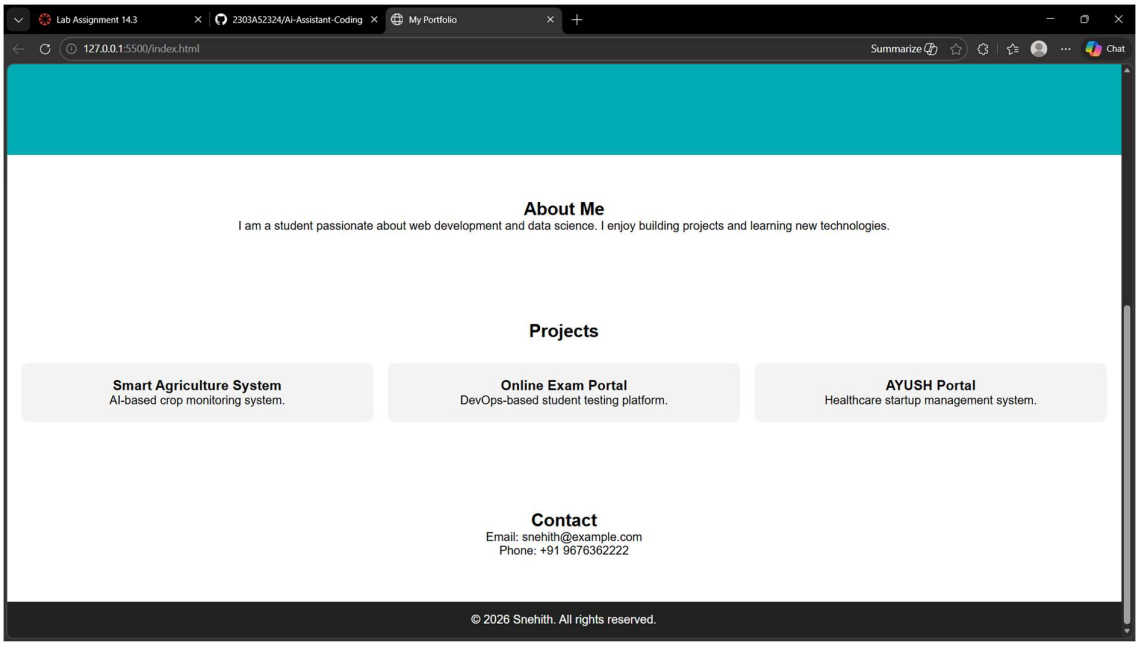
Keep the code simple, well-structured, and beginner-friendly.

Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Portfolio</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<!-- Navigation -->
<header>
    <nav class="navbar">
        <h2 class="logo">MyPortfolio</h2>
        <ul class="nav-links">
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#projects">Projects</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>
</header>

<!-- Home -->
<section id="home" class="section home">
    <h1>Hello, I'm Snehith 👋</h1>
    <p>Aspiring Developer | Data Science Enthusiast</p>
</section>

<!-- About -->
<section id="about" class="section">
    <h2>About Me</h2>
    <p>
        I am a student passionate about web development and data science.
        I enjoy building projects and learning new technologies.
    </p>
</section>
```

Output:





Task 2: AI-Generated Restaurant Landing Page

Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:
    - Home
    - Menu
    - Gallery
    - Contact
2. Generate a menu section styled with CSS cards.
3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.
4. Customize animations or transitions for better user experience.

Expected Outcome

- Fully functional restaurant landing page
- Responsive navigation bar
- Working image slider with smooth transitions

Prompt: Generate a responsive restaurant landing page using HTML, CSS, and JavaScript.

Requirements:

- Create a navigation bar with links: Home, Menu, Gallery, Contact (sticky and responsive using Flexbox)

- Design a Menu section with food items displayed as CSS cards (title, description, price)

- Implement a Gallery section with a JavaScript image slider (auto-slide + next/prev buttons)

- Add smooth animations and transitions (hover effects, fade/slide effects)

- Ensure the website is mobile-friendly (max-width: 768px)

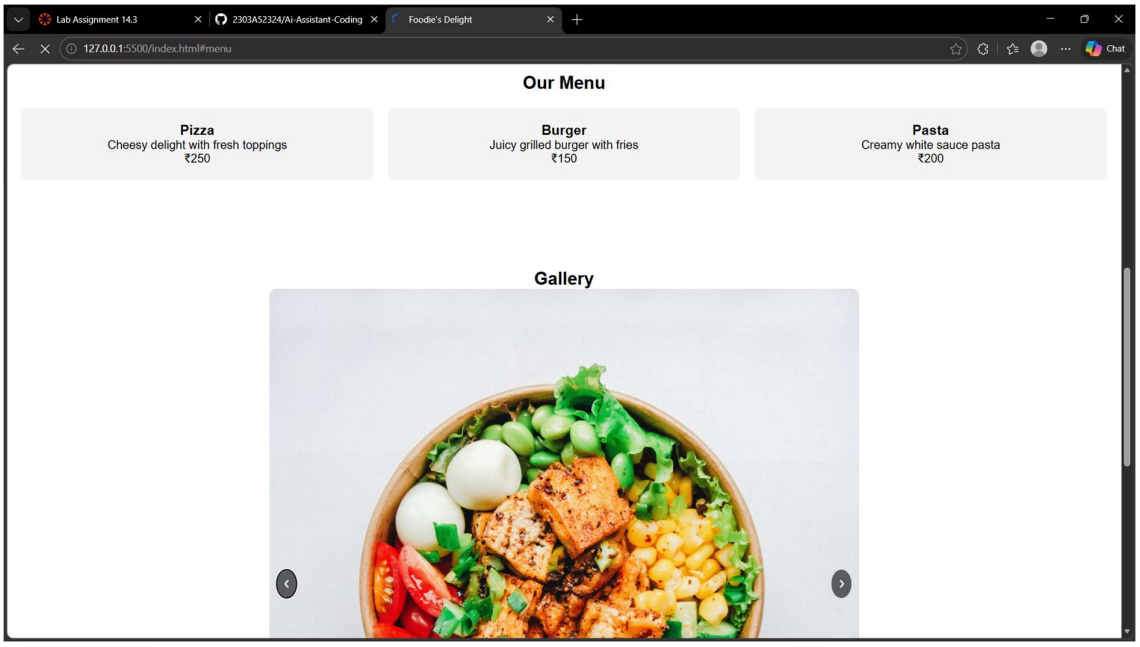
- Use a modern, clean, and attractive design suitable for a restaurant

Keep the code simple, well-structured, and beginner-friendly.

Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Foodie's Delight</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
    <nav class="navbar">
        <h2 class="logo">Foodie's</h2>
        <ul class="nav-links">
            <li><a href="#home">Home</a></li>
            <li><a href="#menu">Menu</a></li>
            <li><a href="#gallery">Gallery</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>
</header>
<section id="home" class="home">
    <h1>Welcome to Foodie's Delight 🍴</h1>
    <p>Delicious food, cozy ambience</p>
</section>
<section id="menu" class="section">
    <h2>Our Menu</h2>
    <div class="menu-container">
        <div class="card">
            <h3>Pizza</h3>
            <p>Cheesy delight with fresh toppings</p>
            <span>₹250</span>
        </div>
        <div class="card">
            <h3>Burger</h3>
            <p>Juicy grilled burger with fries</p>
            <span>₹150</span>
        </div>
        <div class="card">
```

Output:



Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

o Name

o Email

o Phone Number

o Department

o Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.

3. Implement JavaScript validation using Copilot:

o Email format validation

o Phone number length validation

o Required field checks

Expected Outcome

- User-friendly and visually appealing form
- Real-time validation messages

- Prevention of invalid submissions

Prompt: Generate a responsive and user-friendly event registration form using HTML, CSS, and JavaScript.

Requirements:

- Include fields: Name, Email, Phone Number, Department (dropdown), Event Selection (dropdown)

- Use semantic HTML and proper labels for accessibility

- Style the form with modern CSS (centered layout, clean spacing, focus effects, responsive design)

- Ensure the form is fully visible with proper scrolling (no content cut-off)

- Implement real-time validation using JavaScript:

  - Required field checks

  - Email format validation

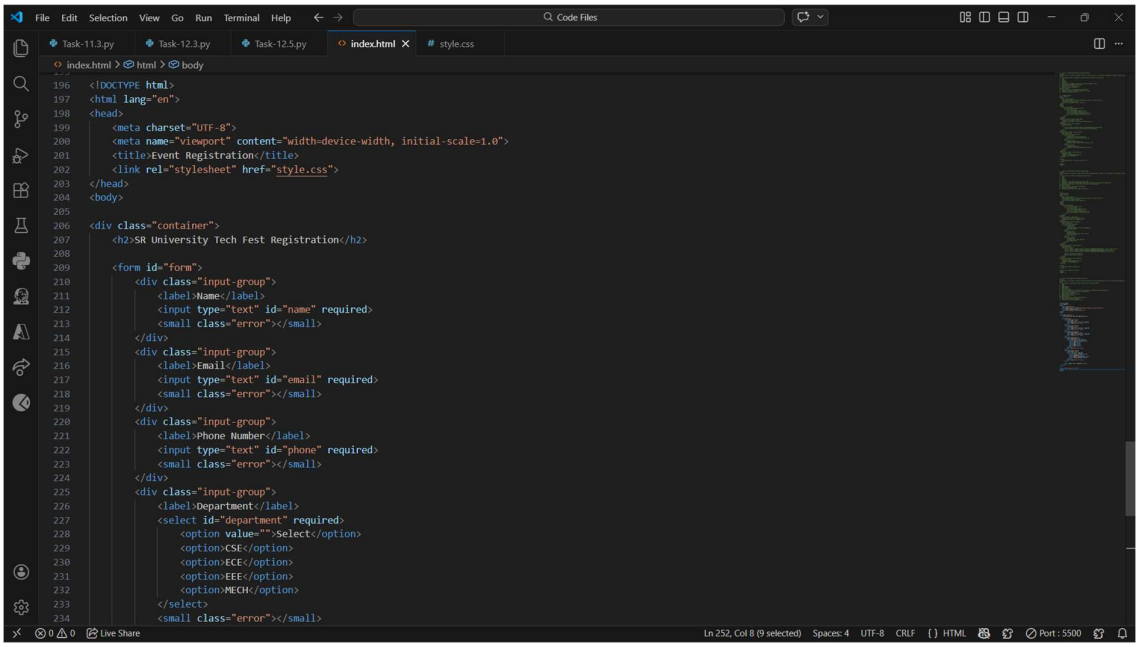
  - Phone number must be 10 digits

- Show error messages below inputs and highlight fields (red for error, green for success)

- Prevent form submission if inputs are invalid

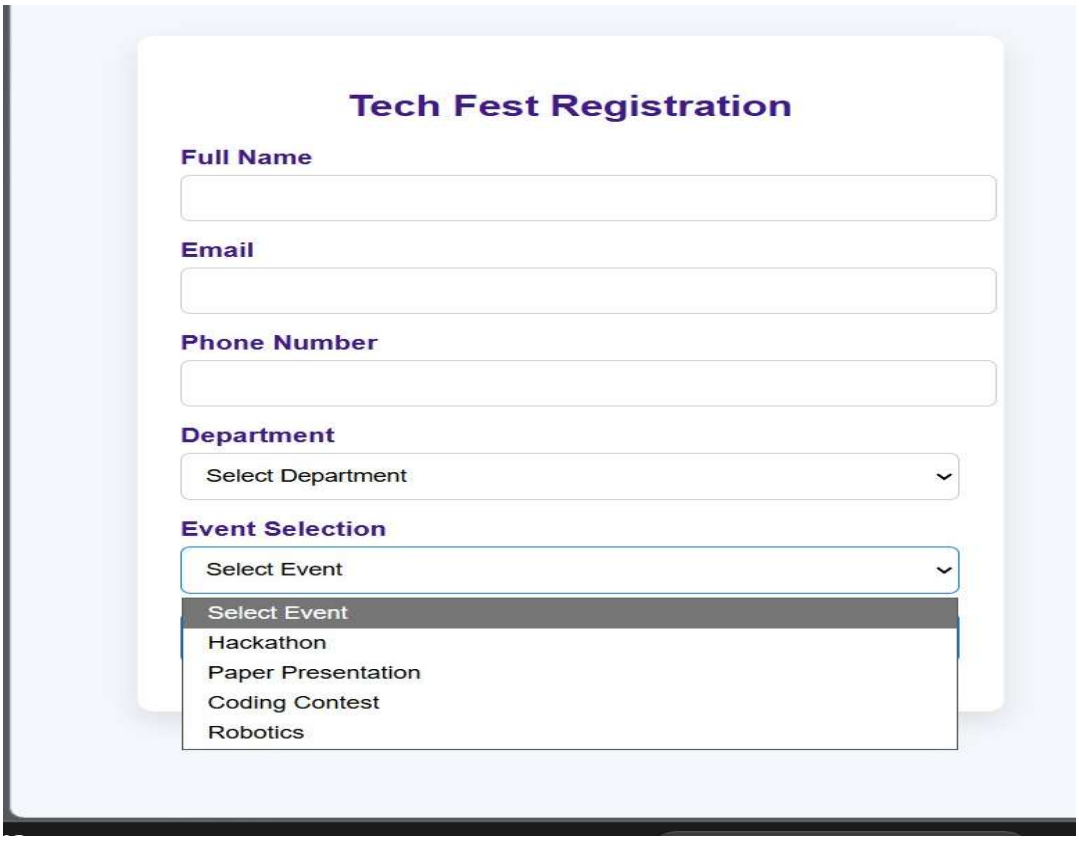
- Keep the code simple, clean, and beginner-friendly

Code:



```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Event Registration</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
    <h2>SR University Tech Fest Registration</h2>

    <form id="form">
        <div class="input-group">
            <label>Name</label>
            <input type="text" id="name" required>
            <small class="error"></small>
        </div>
        <div class="input-group">
            <label>Email</label>
            <input type="text" id="email" required>
            <small class="error"></small>
        </div>
        <div class="input-group">
            <label>Phone Number</label>
            <input type="text" id="phone" required>
            <small class="error"></small>
        </div>
        <div class="input-group">
            <label>Department</label>
            <select id="department" required>
                <option value="">Select</option>
                <option>CSE</option>
                <option>ECE</option>
                <option>EEE</option>
                <option>MECH</option>
            </select>
            <small class="error"></small>
```

Output:



Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an “Add to Cart” feature using JavaScript.
3. Modify AI-generated code to include:
    - Cart item counter at the top-right corner
    - Visual feedback when a product is added

Expected Outcome

- Functional product catalog
- Working cart counter
- Interactive user experience

Prompt: Generate a responsive e-commerce product page using HTML, CSS, and JavaScript.

Requirements:

- Display products in a grid with image, name, price, and "Add to Cart" button

- Add cart functionality using JavaScript

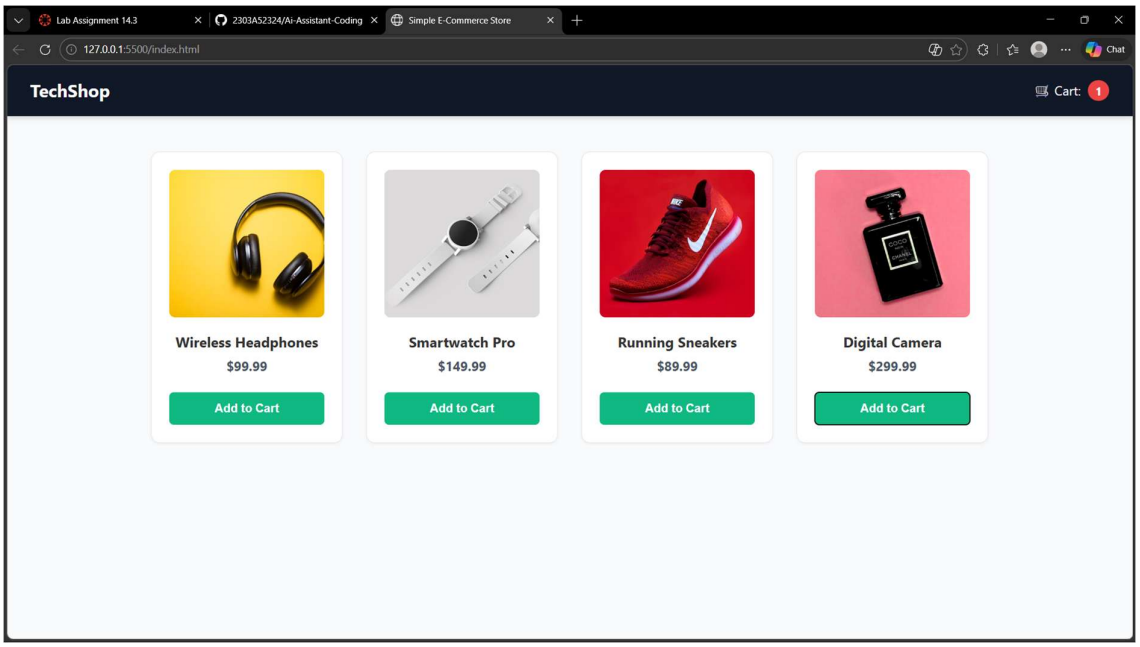
- Show a cart counter at the top-right corner

- Update the counter when items are added

- Provide visual feedback when clicking "Add to Cart" (animation or message)

Use only vanilla JavaScript and keep everything in a single HTML file.

Code:



Task 5: Create a Responsive Web Page Layout

Instructions:

- Design a basic web page layout with a **header, main content area, and footer** using HTML and CSS.
- Use AI to assist in generating **responsive CSS** for different screen sizes.
- Ensure the layout is clean and visually organized.

Expected Output:

- A responsive web page with:
    - Header with navigation links
    - Main section with placeholder text/images

- Footer with copyright or contact info

Layout adapts correctly to desktop, tablet, and mobile screen sizes.

Prompt: Generate a responsive web page layout using HTML and CSS.

Requirements:

- Include a header with navigation links

- Add a main content section with text and placeholder images

- Add a footer with copyright or contact information

- Use CSS (Flexbox or Grid) for layout

- Make the design responsive for desktop, tablet, and mobile using media queries

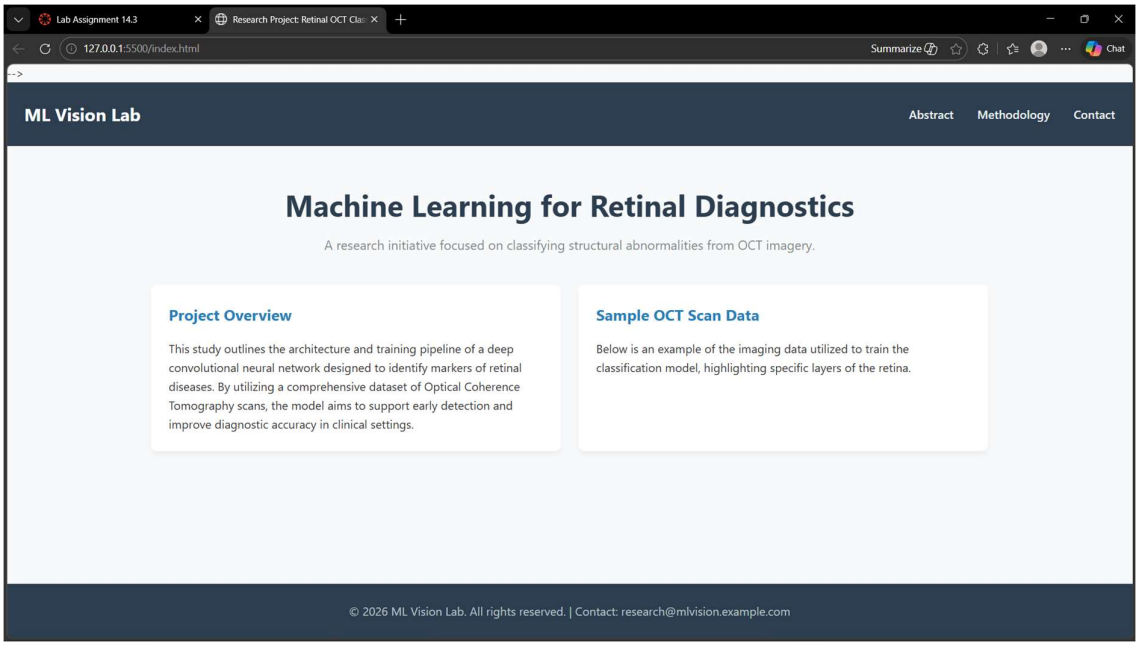
- Keep the design clean and organized

Provide the full code in a single HTML file.

Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Research Project: Retinal OCT Classification</title>
    <style>
        /* ========================================
           BASE STYLES & MOBILE-FIRST LAYOUT
           ======================================== */
        * {
            box-sizing: border-box;
            margin: 0;
            padding: 0;
        }

        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            line-height: 1.6;
            color: #333;
            background-color: #f8f9fa;
            display: flex;
            flex-direction: column;
            min-height: 100vh; /* Ensures footer stays at the bottom */
        }

        /* Header & Navigation (Mobile Stacked) */
        header {
            background-color: #2c3e50;
            color: #ffffff;
            padding: 1.5rem;
            text-align: center;
        }

        .logo {
            font-size: 1.5rem;
            font-weight: bold;
            margin-bottom: 1rem;
```

Output:


Lab Assignment 14.3
Research Project: Retinal OCT Clas
127.0.0.1:5500/index.html
Summarize
Chat
-->
ML Vision Lab
Abstract
Methodology
Contact
Machine Learning for Retinal Diagnostics
A research initiative focused on classifying structural abnormalities from OCT imagery.
Project Overview
This study outlines the architecture and training pipeline of a deep convolutional neural network designed to identify markers of retinal diseases. By utilizing a comprehensive dataset of Optical Coherence Tomography scans, the model aims to support early detection and improve diagnostic accuracy in clinical settings.
Sample OCT Scan Data
Below is an example of the imaging data utilized to train the classification model, highlighting specific layers of the retina.
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